Software Requirements Specification (SRS) for
Luggage Storage Marketplace Application

1. Document Control

- **Project Name:** Luggage Storage Marketplace (Airbnb for Luggage)
- **Version:** v1.0
- **Prepared by:** Ainesh Advani
- **Date:** 15/2/2026
- **Stakeholders:** Founder/Owner, Development Firm, Product Owner, Operations Team, Support Team

2. Goals of this application

- Facilitate customers roam hands-free by finding nearby storage.
- Enable hosts to monetize unused secure space.
- Provide safe, trackable, insured (optional) luggage handover and pickup flows.
- Support reliable payments, refunds, and dispute handling.
- Enable platform that shall be scalable, secure, and provide a reliable "marketplace" experience like Airbnb

3. Objective

To Build a software application (Web application first in MVP, Mobile Application later) that allows below mentioned roles to perform following functions:

1. **Service Providers (Hosts)** to list and manage storage spaces (home/shop/office/warehouse/hotel) and accept luggage storage bookings, manages check-in/check-out, receives payouts, reviews/rates end users

2. **End Users (Guests)** to discover and search storage locations, book storage for minimum of **1 day to 365 days**, pay securely, and drop/pick luggage using a verified process, rates provider

3. **Admin (Platform Administrator)** Verifies providers, manages disputes/refunds, monitors safety/fraud, configures fees, runs reports.

4. Scope

Scope of the application is categorized into two parts.

- First part captures In Scope (MVP) which shall be implemented as Phase 0 as part of Web Application.
- Second part captures the Out of scope from present MVP and shall be implemented in later phases.

4.1. In Scope (MVP) – Web application

Core MVP value will be to enable following functionalities

- **Seamless Onboarding & Listing**: User registration/login (Guest/Host) & authentication; Host onboarding & Verification; Facilitate listing creation by Host; Facilitate User registration as an end user or as a Service Provider or both but using different flows
- **Discovery**: Location-based search (google maps + list), and Availability-based on dates/time (initial booking will be for whole day)
- **Pricing & Booking**: Price per bag per day, overbooking prevention, reflect booking states.
- **Trust and safety:** Reviews/Ratings, Basic KYC/ID capture (manual review in MVP), Secure and Verified drop-off/pickup (OTP/QR) as part of Check-in/check-out.
- **Payments & Refunds**: Online payments (Stripe globally / Razorpay for India) with configurable refunds. Platform fees, Hosts payouts, Fixed pricing initially set beforehand, Compliance to Privacy and payment security best practices (PCI by payment gateway)
- **Operations & Customer Care:** Admin panel for user/listing/booking oversight, payouts ledger to hosts; Support (Incident management) flow
- **Communication**: Messaging/Notification, In-app chat

4.2. Out of Scope for present MVP (Below tasks shall be handled in Phase 1 and beyond)

- **Pricing & Booking**: Overbooking prevention (Capacity locking for selected time window (with timeout)) , Auto disabling of storage location from search once capacity is full, Landmark proximity (Filter Property Proximity to key areas), Pricing Control with host, 3D Image of Host space and capacity management, OTP based Auth to be implemented later and other social logins like apple, Facebook etc.,
- **Trust and Safety**: Smart locker, Insurance claim handling workflows, Full KYC/ID verification automation, Luggage Tag Code, AI fraud detection
- **Operations and Customer Care**: Loyalty program/ coupons/ multi-day discounts, Host Listings verification, Multi-city corporate partnerships API, Multi language support apart from English, Must support time zones and local currency (configurable), Develop mobile app, Price review by admin and approval on edits, Pickup luggage from user place
- **Payments & Refunds**: Advanced dynamic pricing per host determined by factors like duration, season, and host preferences, Update Refund rules as per user feedback
- **Communication:** Booking confirmation additional channels like Whatsapp, SMS and association notifications preference

5. Assumptions & Constraints

- Host KYC verification will be done by admin
- Guest & Host must be 18 yrs and above.
- Guest KYC verification will be done by host on the time of drop-off
- Drop-off/pickup should be verifiable (OTP/Code)
- Luggage Storage duration range: 1 day to 365 days
- Location services required (Google Maps + GPS to determine current location)
- Guests must comply with platforms' and hosts' luggage rules and regulations (including prohibited items)
- Pricing will be per bag per day (day counted every 24 hours and it starts from the start of luggage drop-off time. Partial days will be rounded up to next day), with optional minimum charges
- Platform commission model: percentage (40% platform provider and Host 60%) or fixed fee (configurable)
- Currency used will be INR, USD, EURO.
- Date/Time, Number formatting will be Localized as per the listing country; For example time zone always explicit (IST) when operating in India

6. User Types, Roles and Journeys

6.1. Guest (End User)

- **Summary of Journey -** Discover → Select → Book → Pay → Drop-off (check-in) → Pickup (check-out) → Rate/Review -> Raise support request (if necessary)
    - Select city / use current location/ browse on map
    - Search storage (map + list)
    - View listing details (price/bag/day, operating hours, reviews, restrictions)
    - Select drop and pick up time (auto determine the no. of days)
    - Select number of bags and size of bags to drop off
    - Initiate Booking, wait for host confirmation and account to be verified
    - Make payment if already logged in or post sign-up and once booking is confirmed by host
    - Receive booking confirmation through email+ drop-off code
    - Receive directions link to storage location via google maps
    - Drop luggage → Host verifies through OTP/Code
    - Pick up luggage → Host verifies through OTP/Code
    - Invoice receipt
    - Rate/review host
    - Raise support requests (if necessary)

6.2. Host (Service Provider)

- **Summary of Journey -** Onboard/KYC → List location → Set schedule/capacity/prices (phase 1 & beyond) → Manage bookings → Check-in/out → Payout → Reviews/Rates
    - Signup/login as Host
    - Create profile
    - Verifies themselves by sharing personal information (Legal Name, DOB, country, Government ID type, Government ID no., Selfie)
    - Creates listings and update following information e.g address, photos of the facility, luggage capacity per type (small, medium, large), rules, open or closed, timings (opening and closing hours), pricing (phase 1 & beyond), safety features checklist e.g CCTV, staff presence, locker room, Availability of luggage storage facility through availability calendar
    - Share bank account details
    - Receives and manages bookings
        - Receive booking request – either approve or cancel booking request with reason
    - Confirms drop-off/pickup (verification process)
        - At drop-off:

        - Collects guest's government id (passport, Aadhaar, PAN), Verifies the guest using govt. documents.
        - Take photos of Govt. Id and luggage and upload it to this app to confirm drop-off
        - Verify booking code, mark "checked-in"
    - At pickup
        - Verifies the guest using govt. documents.
        - Verify code, mark "completed"
        - Take photos of luggage and upload it to this app to confirm pick-up
- Manages operating hours, storage rules at predetermined pricing
- View earnings dashboard, request payout
- Reviews/Rates Host

6.3. Admin (Platform Operator)

- **Summary of Journey -** Verify → Configure and Administrate policies → Reviews and Approve the Listings by Service Provider -> Handle incidents → Monitor KPIs → Finance reconciliation

    - Verifies host (KYC-Manual)
    - Approves/Rejects listings with Reasons
    - Overlooks listings edits (approve, reject or put on hold)
    - Listings will be visible only post approval
    - Handles disputes, refunds, chargebacks
    - Manage users, content moderation
    - Monitors fraud, suspicious activity
    - Manages commissions, payouts configuration
    - Monitors Reviews

6.4. Support Agent

- Handles tickets, escalations

7. Functional Requirements

7.1. Authentication & Account Creation

- FR- 1 User signup via Google OAuth via Firebase
- FR- 2 User can register as Guest or Host (or both but with separate flows)
- FR- 3 Profile management - Name, Phone number, Alternate phone number, Email, Photo, DOB
- FR- 4 Device/Session management.

7.2. Account Verification

- FR- 5 Host to submit the PII/KYC details
- FR- 6 Administrator to verify the PII/KYC details and notify the Host

7.3. Host Onboarding & Listings

- FR- 7 Create listing with
  - Property Type (shop, hotel, house, warehouse etc.)
  - Listing title, description (Initially auto set this: Title as User's <property type>, description)
  - Full address + Map pin
  - Photos (initially just 2 photos with ability to select one as main photo)
  - Storage capacity for each size: Small (tote bags, purse), Regular (suitcases, backpacks), Large (odd-sized goods like cartons, cycle, golf bags)
  - Allowed items / Prohibited items - hosts can refuse luggage storage if it contains prohibited items on arrival.
  - Operating hours and days. Option to blackout non-operational days
  - Safety / Additional features (CCTV, Staff present, WiFi, Restrooms)

- FR- 8 Listing status: Draft → Submit -> Pending Review → Approved/Not Approved -> Active/Inactive (send notification)

- FR- 9 Listing availability management:
  - Set weekly schedule + exceptions (dates) (Calendar view)
  - Block times / Time windows
  - Black out days/Holidays
  - Price Per bag per day (Initially set by platform in MVP and later we can give flexibility to host to set price but not go below threshold)
  - Minimum booking duration
  - Multi-day discounts
  - Platform commission settings (admin-side)
  - Reviews
  - Cancellation Policy
  - Directions (Google/Apple Maps integration)

7.4. Search & Discovery (Guest)

- FR- 10 Guest search by:
  - Current location / city / landmark/ Map
  - Allow GPS location permission to facilitate current location search
  - Request customer location enablement
  - Show users location on map so they can see which location is closest to them
  - Date/time range/open now
- FR- 11 Results show on map + list:
  - Price, Rating, open/closed based on host operating hours relative to listing's local time zone, Property Listing
- FR- 12 Filters:
  - Rating, Open now,Storage type, Security features (optional), Restroom, Wifi
- FR- 13 Listing detail page shows:
  - Property details, photos, availability, perks of using our app
  - Total price breakdown
  - Prohibited items: Checkbox confirmation + policy link at checkout; host can refuse prohibited items on arrival.
  - Cancellation/refund policy
  - Reviews

7.5. Booking & Reservations

- FR- 14 Guest selections:
  - Start/end date-time (Drop off and Pickup time)
  - Number of bags
  - Bag Size
- FR- 15 System calculates pricing:
  - Bag count × time × rate
  - Taxes/fees/commission (line item)
  - Final Price inclusive of Taxes
- FR- 16 Payment page
  - Various modes of payment: UPI, Google Pay, Card
  - Acceptance of terms and conditions (hyperlink for guest to view)
- FR- 17 Booking states:
  - Pending Host Confirmation
    - Auto-accept (recommended) OR manual accept within 6 hours (configurable)
    - Booking request will get auto rejected after 6 hours
  - Pending Payment → Confirmed
  - Checked-in (after drop-off)
  - Completed (after pickup)

- Host Not Available
- Cancelled
- No-show
- Initiate Refund (as applicable)
- Refunded (as applicable)

▪ FR- 18 Booking confirmation after payment includes:
- Booking ID
- Host address + navigation link
- Drop-off instructions
- Secure verification code (OTP/QR)

▪ FR- 19 Booking modifications:
- Allow time extension if capacity allows
- Pricing difference collected/refunded automatically based on rules

7.6. Payments, Refunds, Payouts

▪ FR- 20 Integrate payment gateway(s) (e.g., Razorpay for India/Stripe for international).

▪ FR- 21 Idempotency key on payment/booking create to prevent duplicates. Retry with idempotency keys.

▪ FR- 22 Compliance to 3-DS flows

▪ FR- 23 Payment statuses: initiated/success/failed/refunded.

▪ FR- 24 Refund rules configurable by admin:
- Full refund if cancelled 1 day before booking

▪ FR- 25 Host payout handling:
- Bank account, Payout setup
- Wallet ledger (earnings, refunds)
- Payout schedule: monthly (configurable) and after the completion of booking
- Exportable statements (CSV/PDF)
- Payout cycles payments (no card storage in app backend)

7.7. Drop-off & Pickup Verification

▪ FR- 26 At drop-off, host verifies booking using:
- OTP shown in guest app OR booking code
- Verifies user via their government documents
- Checks luggage for prohibited items

- Uploads luggage photos, remarks about luggage condition and guest government documents on our portal

▪ FR- 27 On verification, host marks booking as Checked in.

▪ FR- 28 Host confirms number of bags and bag sizes

▪ FR- 29 Host optionally takes bag photos / notes (condition)

▪ FR- 30 At pickup, host verify again using OTP/Checkout Code, check government id, upload luggage photos, conditions and mark checkout completed.

▪ FR- 31 Late pickup handling

- If pickup time exceeded:
  - System applies late fee rules (configurable)
  - Notifies customer
  - Provider dashboard shows "Overdue"

7.8. Ratings, Reviews, Support

▪ FR- 32 Guest can rate (1-5) host/listing after completion with optional comments

▪ FR- 33 Host can also rate and review the Guest

▪ FR- 34 Review moderation tools for admin.

▪ FR- 35 Trust badges:

- "Verified Provider"
- "CCTV Available"
- "High Rating"
- "Fast Check-in"

▪ FR- 36 In-app support:

- FAQ page
- Support ticket submission with booking reference
- Admin/support response workflow (email integration initially)

7.9. Notifications

▪ FR- 37 Email notifications for: (later we can add SMS, Whatsapp channels)

- Booking Request
- Booking confirmation
- Payment confirmation to end user
- Payout confirmation to Host
- Reminder before drop-off/pickup
- Cancellation/refund updates
- Check-in/out updates

- Overdue pickup
- Refund Issued/Processed
- Admin pushing notifications

▪ FR- 38 Messaging In-app chat:

- End user ↔ Host after booking
- Mask phone numbers (no direct sharing by default)

7.10. Terms, Conditions and Disclaimers

▪ FR- 39 Host – Host(s) should list only those listings that belong to them

▪ FR- 40 Guest – Carry govt. id with you when dropping off luggage

▪ FR- 41 Refund policy" link near Confirm Booking Button.

▪ FR- 42 Guest(s) represents and warrants that all luggage and items deposited for storage belong to them

7.11. Admin Panel (Back Office)

▪ FR- 43 Admin dashboard: KPIs (bookings, revenue, disputes, ratings).

▪ FR- 44 Manage users/hosts: Verify – Approve/Not Approve with reason; Suspend/Blacklist; View/search customers/providers; Reset verification status; Audit log; Document review

▪ FR- 45 Manage listings: Approve/Reject, Edit, Deactivate, Flag suspicious listings

▪ FR- 46 Manage bookings: Cancel in unusual circumstances, Refund, resolve disputes.

▪ FR- 47 Manage commissions, taxes, policies, coupons.

▪ FR- 48 Audit logs for critical actions with user id, timestamp, session id etc.

▪ FR- 49 Financial Controls

- Configure commission/service fees
- Taxes/fee rules by city/country
- Refund overrides
- Payout batch management
- Export reports (CSV/PDF)

7.12. Edge Case Scenarios

▪ FR- 50 If booking has to be extended or cut short after a luggage has been cheeked-in

- In case of change in time period (extension or reduction) – User shall be able to select extend or reduce the time period. Accordingly, pricing will be calculated, and payment or refund request will be initiated

7.13. Disputes, Support & Incident Handling

- FR- 51 Dispute types:
  - Damaged item
  - Missing item
  - Provider unavailable
  - Wrong charges / refund issue

- FR- 52 Dispute console:
  - Assign dispute to agent
  - Facilitate guest/host to upload photos
  - View chat logs, check-in/out evidence, photos, booking event life cycle
  - Apply adjustments/refunds

- FR- 53 Support:
  - In-app support ticket creation with attachments
  - Status tracking
  - Admin resolution workflow

8. Non-Functional Requirements

8.1. Performance

- FR- 54 Search results must load within <1 seconds under normal load.
- FR- 55 Whole website should be dynamic and reactive and work on all screen sizes
- FR- 56 Booking flow must be reliable with transactional integrity (indexing, no double booking by same user).
- FR- 57 Session timeout and extension functionality shall be implemented (session duration 5 min)
- FR- 58 **Map/geo performance**: Use **geohash** or PostGIS index (GIST) on coordinates; pre-compute "open now" flags for current day/hour.

8.2. Scalability

- FR- 59 Must support multi-city expansion.
- FR- 60 Architecture should allow scaling API + database + search.

8.3. Availability & Reliability

- FR- 61 99.9999% uptime target (MVP), higher later.
- FR- 62 Automated backups and disaster recovery plan.

8.4. Security

- FR- 63 Encryption in transit (TLS) for sensitive data.
- FR- 64 Secure payment handling (PCI compliance via gateway).
- FR- 65 Role-based access control (Guest/Host/Admin).
- FR- 66 Rate limiting, anti-bot on login/OTP endpoints.
- FR- 67 Logging + monitoring for suspicious activity.
- FR- 68 Vulnerability and Penetration testing
- FR- 69 Restricting multiple attempts e.g Booking, Editing etc.
- FR- 70 OWASP Top 10 protections

8.5. Privacy & Compliance

- FR- 71 Comply with applicable data protection laws (India DPDP / GDPR if global).
- FR- 72 Clear data retention and deletion process.
- FR- 73 Consent for location permissions.

8.6. Maintainability

- FR- 74 Clean modular code, documented APIs.
- FR- 75 Automated tests for booking/payment flows.
- FR- 76 CI/CD pipeline for deployment.

8.7. Audit & Logs

- FR- 77 Action Logs – Admin/User/Host actions log
- FR- 78 Booking lifecycle events log – starting from booking till cancellation or completion
- FR- 79 Every Transaction must be tracked through session id, and all actions are tagged against that

9. Other Key Requirements

9.1. Data Requirements - Core Entities

Below is just a example of the DB schema. Updates shall be made as needed

- FR- 80 User: id, role(s), name, email, phone, status-active/inactive, createdAt
- FR- 81 HostProfile: userId, verificationStatus, payoutDetails, address, status-active/inactive, createdAt
- FR- 82 Listing: id, hostId, title, address, geoLocation, photos, rules, status
- FR- 83 Availability: listingId, schedule, exceptions, capacity
- FR- 84 Booking: id, guestId, listingId, start/end, bagCount, price, status
- FR- 85 Payment: bookingId, gatewayRef, amount, status, timestamps
- FR- 86 Payout: hostId, amount, status, bankRef, period
- FR- 87 Review: bookingId, rating, comment, moderationStatus
- FR- 88 SupportTicket: userId, bookingId (optional), message, status

9.2. Integrations

- FR- 89 Maps: Google Maps (geocoding, map pins, distance)
- FR- 90 Payments: Razorpay (India) / Stripe (global)
- FR- 91 Notifications: SMS (Twilio or local provider), Email (SendGrid), Push (Firebase)
- FR- 92 Analytics: GA4/Mixpanel
- FR- 93 Storage: Cloud storage for images (S3 / GCS)

9.3. UI/UX Requirements

- FR- 94 Guest App Screens shall have
  - Onboarding/Login
  - Home/Search + map
  - Filters + listing cards
  - Listing details
  - Booking checkout + payment
  - My bookings (upcoming/completed)
  - Booking details with drop-off/pickup code
  - Profile, support, reviews
- FR- 95 Host App Screens shall have
  - Host dashboard
  - Create/edit listing

- Availability calendar
- Booking requests and schedule
- Verify drop-off/pickup screen
- Earnings & payouts
- Profile settings
- Comment on Guest Reviews and Review the Guest

▪ FR- 96 Admin Web Panel shall have

- Dashboard
- User (Host/Guest) management
- Listing management
- Booking/payment management
- Dispute/support management
- Configurations/updates

9.4. Business Rules (Examples)

▪ FR- 97 Minimum booking duration: 1 day

▪ FR- 98 Maximum booking duration: 365 days

▪ FR- 99 Multiple bookings check

▪ FR- 100 Listings don't show deactivated properties

▪ FR- 101 Cancellation and refund policy applies based on the remaining time. Cancellations: Full refund ≥ 24h prior to start; otherwise, 0% (admin-configurable).

▪ FR- 102 Host can toggle "instant booking" vs "manual approval"

▪ FR- 103 Prohibited items list enforced during booking confirmation

10. Testing & Acceptance Criteria

10.1. Must-Pass Scenarios

- Search → Initiate Booking -> ID verification & Host approval → Initiate Pay → Drop-off verified → Pickup verified → Review
- Overbooking prevention under concurrent bookings
- Payment workflow
- Refund workflow works end-to-end
- Admin can disable a listing, and delisted listing shall disappear from search
- Host payout ledger shall reconcile after cancellations/refunds

10.2. Testing Types

- Unit testing (backend, frontend + apps)
- API testing
- Integration testing (payment, maps, e2e call flows, audit trail verification)
- UAT (end-to-end journeys)
- Security testing (including VAPT)
- Performance testing for end-to-end call flow

11. Key Deliverables

- Documentation
  - Requirement Specifications
  - Design - HLD and LLD covering User flows
  - Testing – Testcases and results for SIT, UAT, Performance Testing, Non-Functional testing
  - Installation and Configuration Manual
  - Operations Document
  - User Manual
- Software having
  - Backend APIs, Swagger/Postman collection
  - Web Application
  - Scripts for Deployment setup + environment configuration
  - Source code repository
- Post-go-live support plan and SLAs